

GBIF occurrence taxonomy harmonization workflow

Vertical clean version with bridge detail and roll-up example.

INPUT TABLES

Cleaned occurrence
zosteropidae_gbif_clean.tsv
GBIF accepted_name_gbif + accepted_key_gbif

BirdLife/HBW reference
birdlife_reference.tsv
Species, subspecies and synonym names

Avibase/eBird reference
avibase_reference.tsv
Clements/eBird + Avibase names linked to HBW/BirdLife code

Manual override table
manual_taxonomy_overrides.tsv
Reviewed taxonomy cases only

MAIN FLOW

Extract unique GBIF species
Collapse occurrence rows to one row per GBIF species concept. Count records per accepted_name_gbif + accepted_key_gbif.

1. Direct BirdLife/HBW match
Search GBIF name key in BirdLife species names, subspecies names and synonyms.

One BirdLife target?

2. Avibase/eBird bridge
Used when direct BirdLife matching fails. It searches the GBIF name in Avibase/eBird names, then follows HBW/BirdLife code back to the BirdLife species concept.

Bridge detail
GBIF name -> search_name in avibase_reference -> avibase_id/name_id -> hbw_birdlife_name_code -> BirdLife species SISRecID.

One BirdLife target?

3. Parent-species roll-up
For split/lump cases: look at subspecies rows where parent_species_name equals the unmatched GBIF species. If they point to one BirdLife species, accept it.

Roll-up example
GBIF: Zosterops anderssoni
Avibase subspecies rows point to Zosterops senegalensis stierlingi/tongensis/anderssoni.
Result: BirdLife species = Zosterops senegalensis.

All linked subspecies point to one BirdLife species?

4. Manual override
Used only after review. Example: GBIF Zosterops montanus -> BirdLife Zosterops japonicus in manual_taxonomy_overrides.tsv.

RESULT IF CONDITION IS TRUE

Yes -> matched_birdlife_reference
Direct BirdLife/HBW match.

Yes -> matched_avibase_bridge
Exact Avibase/eBird bridge match.

Yes -> matched_avibase_bridge
Parent-species roll-up accepted.

Multiple -> ambiguous_review
Several possible BirdLife targets; keep for review.

Reviewed row -> matched_manual_override
Manual taxonomy case accepted.

No row -> no_match
No target found automatically or manually.

Final output files
1. gbif_to_birdlife_species_crosswalk.tsv - species-level GBIF to BirdLife/HBW mapping
2. zosteropidae_gbif_clean_birdlife_harmonized.tsv - all occurrence rows with harmonized taxonomy fields
3. zosteropidae_gbif_clean_birdlife_matched.tsv - matched occurrence rows only
4. summary_gbif_to_birdlife_taxonomy.tsv - counts by match status

Manual override rows are stored outside the R script so future reviewed cases can be added without changing the code.
Ambiguous and no_match species are retained in the crosswalk for review; this taxonomy step does not delete occurrence records.
Priority order: direct BirdLife/HBW match > Avibase/eBird bridge > manual override.